Compound Degradations in Maritime Ship Recognition: a Paired Protocol, an Error-Union Decomposition, and Coverage of Co-occurrence as What Removes the Excess

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Abstract

Maritime images are rarely degraded by one factor at a time: haze, cloud cover and sun glare co-occur, and operational recognition systems meet them in combination. Existing maritime robustness studies nonetheless vary one degradation at a time. We introduce a paired evaluation protocol for compound degradations and apply it to fine-grained ship-type recognition on FGSC-23 (eight civilian classes), a protocol that fixes the preprocessing pipeline, derives the corruption instance for every image from a deterministic function of image identity, and evaluates every condition per image, so that any two models — or any two conditions — can be differenced without confounding. Four backbone families show compound conditions dropping accuracy by up to 43 points, non-linearly in the single-factor drops. A pre-registered four-point severity design locates the non-separability: at severities 2–4 the compound drop is statistically indistinguishable from its additive prediction, whereas at severity 5 it exceeds that prediction by +11.3 pp (95% CI [+7.2, +15.3], BH $q < 0.001$) and exceeds the largest single-degradation drop in 6 of 6 runs. Against the stronger independent-failure baseline — the union of the constituents' error sets — 41.5% of severe-compound errors are made by no single degradation at any severity, so they are not harder versions of the same errors.

We then ask what removes the excess, and report a positive result with a measured boundary. Training on a pool of calibrated compound samples drawn from six families reduces the excess on three pairs held out of every pool from +8.2 pp to −1.8 pp (clean accuracy 84.95% → 86.25%), and a control that shows the model the same six families one at a time, at the same compound rate and budget, stays at +7.2 pp: the paired-minus-singles difference is −5.8 pp, 6 of 6 cells. A third arm matched on damage per image recovers about 60% of the gain, so the honest accounting is roughly 60% under-matched damage and 40% co-occurrence; the co-occurrence part survives matching (−3.7 pp, 6 of 6) and is the part that transfers to unseen pairs. Neither part transfers to families absent from all pools, as pre-registered. Eight single-point interventions (seven mechanisms plus an initialization control) fail to improve on a strong combined-augmentation baseline, and we document one methodological artefact that matters beyond this task: a +1.66 pp gain measured at a 24-epoch screening budget becomes −0.18 pp at 40 epochs. We release the protocol, the harness, all checkpoints and a manifest that reproduces every number.

Keywords: maritime computer vision; ship classification; robustness evaluation; compound degradations; superadditivity; negative results

1. Introduction

A ship photographed from shore in the Yellow Sea is not degraded by either haze or glare; it is degraded by some mixture of both, on top of an unknown amount of cloud. Recognition models deployed on such imagery are therefore evaluated under conditions that no single-corruption benchmark represents.

The published maritime literature has moved in the opposite direction. FoggySea isolates sea fog [11]. CARP isolates cloud occlusion [12]. Studies of haze on warship classification isolate haze. Benchmarks that do consider multiple corruptions — ImageNet-C [3] and its descendants — hold the number of types at one and vary only severity. Recent general-vision work studies composition, but typically at the level of "two or more corruptions applied in sequence", without decomposing the joint effect against the marginal effects of its parts. As far as we can establish, no maritime study reports how a compound degradation compares to its constituents, and no study in any domain reports an explicit factorial decomposition of accuracy loss on ship-type recognition.

We therefore ask a narrower question than "how robust are ship classifiers": what does a compound degradation do that its constituents do not, and how does that difference depend on severity?

Answering it required three design decisions that turned out to matter more than any model:

1. Paired corruption. The corruption parameters for an image are drawn from stable_seed(recipe_version, sample_id, role, kind, severity, domain). Two models evaluated on the same condition see the identical corrupted image; two conditions evaluated on the same model differ only in the degradation. This turns every model comparison into a paired comparison and removes most of the seed noise that otherwise dominates at 157 validation images.
2. Pre-registered screening with full-budget confirmation. Screening statistics were fixed before training. The protocol's value is shown by the fact that it caught its own artefact: a candidate that looked like +1.66 pp at a 24-epoch budget was −0.18 pp at 40 epochs.
3. Analysis of stored predictions, not just aggregates. Every condition's per-image prediction is archived, which lets us ask questions that aggregate accuracies cannot answer — in particular whether compound failures are the same failures a single degradation makes.

Contributions.

- A paired protocol for compound-degradation evaluation in maritime recognition, with a strong combined-augmentation baseline, group-disjoint folds and cluster bootstrap inference (§3).
- Evidence that the effect of compound degradation is not separable into its constituents, and that under the shipped renderer this non-separability appears as a threshold confined to the highest severity rather than a smooth gradient (§4.3) — including one pre-registered prediction that failed. A second, physically-motivated renderer, applied both to a frozen model and to models retrained under it, reproduces the severity-5 excess (10.2 / 16.0 pp against 9.7 / 12.6 pp) but produces it from severity 4 upwards: the excess is not an artefact of one rendering, while the onset severity is (§8.4).
- Evidence that compound failures are qualitatively new: 41.5% of severe-compound errors occur on images that every single degradation leaves correct (§4.4). Against the stronger independent-failure baseline (the union of single-corruption error sets), the severity-5 haze+glare excess survives in 6/6 cells (+7.3 to +13.1 pp, mean +9.9), while the haze+cloud excess vanishes (mean −0.7 pp); at severities 2–4 both pairs are mildly sub-additive (§4.7).
- A composition study over fifteen calibrated pairs: at matched marginal damage the interaction decomposes additively into per-family main effects (R^2 0.42–0.90 across two backbones and twelve cells), no single-corruption geometric predictor recovers the ordering, and the per-family values themselves are backbone-dependent (§4.7).
- Evidence that the excess is what an uncovered operating point looks like, and that it is removable: a pool of calibrated compound samples cuts the held-out excess from +8.2 pp to −1.8 pp (6/6 cells), the same six families applied one at a time do not (−5.8 pp paired difference, 6/6), roughly 40% of the gain survives an arm matched on per-image damage (−3.7 pp, 6/6), and the effect transfers to unseen pairs but not to unseen families (§5).
- A pre-registered negative result on eight interventions, and a documented case in which a limited training budget manufactures a gain (§6).

2. Related work

Maritime datasets and degradation studies. FGSC-23 [13] provides fine-grained ship classes; the civilian subset used here has eight classes. FoggySea [11] benchmarks detectors on synthetically fogged SeaShips. CARP [12] studies few-shot ship classification under cloud occlusion at controlled coverage. A recent study of FGSC-23 under synthetic cloud [6] reports uncertainty behaviour, not compound effects. Work on haze and warship classification reports monotone accuracy decay with fog density [11]. All of these vary one factor.

Corruption benchmarks. ImageNet-C [3] established 15 corruption types × 5 severities with the mean corruption error (mCE) as the summary statistic; by construction it never combines types. ImageNet-3DCC [4] renders physically motivated corruptions but still one at a time. Mintun et al. [8] showed that augmentation-derived robustness is corruption-specific and that mCE can be gamed by similarity — a warning we take seriously and address by never tuning augmentation to the evaluation corruptions.

Composition. Two-factor perturbation studies [1] report that a second factor sometimes helps. Compositional robustness work [7] stacks 2–6 corruptions and argues invariance does not transfer. Decision-boundary analyses of combined corruptions [10] examine which combinations break robustness. None of these decomposes the joint effect against the product of marginals, which is precisely the quantity that distinguishes "harder" from "different".

Negative results as method. DomainBed [2] and RDumb [9] established that careful, well-controlled benchmarks often find simple baselines unbeaten. We follow that tradition rather than claiming an algorithmic contribution we do not have.

3. Protocol

Data. FGSC-23 [13], civilian scope, eight classes. 618 training and 157 validation images. Three group-disjoint folds over the training set (412/206 per fold), split by source_group_id so that no vessel or scene appears in both sides. The 236 images in the dataset's test split are not used for any result in this paper and are disclosed as compromised as a blind set (§8.3).

Preprocessing. Frozen: ImageOps.contain(image, (224, 224), BICUBIC) → corruption → ImageOps.pad((224, 224), color=(124, 116, 104)) → ImageNet normalization (mean [.485, .456, .406], std [.229, .224, .225]). No augmentation touches evaluation images.

Corruptions. Three physical families — haze (atmospheric transmission with random airlight), cloud (soft spatial support with alpha blending), glare (Gaussian bloom with gain) — each at five severities, implemented in degradations.py and frozen with a recipe hash recorded in every run. Compound kinds apply both components with independent random fields. Two domains: matched (fields drawn at the same scale as the scene) and shifted (fields drawn at a different scale, a stricter domain-shift control). 43 conditions are evaluated per model: clean, 30 single, 12 compound, plus one placement diagnostic.

Training. ConvNeXt-T [5] (torchvision ImageNet-1K weights, 27.8M parameters), batch 32, 40 epochs, cosine schedule, and a combined augmentation recipe that samples clean in 25% of batches, single degradations in 50% and compounds in 25%, with severities drawn from the recipe's own distribution. Two seeds (17, 29) per fold.

Inference. Cluster bootstrap over source_group_id, 2000–4000 resamples, with the two seeds of an image kept in the same cluster. Multiplicity controlled by Benjamini–Hochberg where a family of tests is reported. All gains are computed per image, same fold, same seed, same corruption instance.

Total cost. 136 training runs, ≈3.5 GPU-hours on a single RTX 4090.

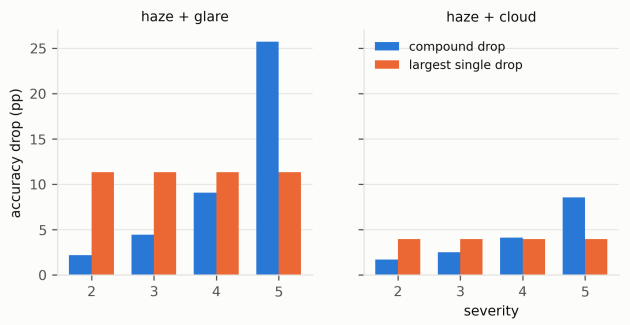


Figure 1. Accuracy drop (pp) of each compound condition against the largest drop produced by any severity of either constituent, baseline arm, mean over 3 folds × 2 seeds.

4. Results

4.1. Single degradations are unequal, and haze is nearly harmless

On the baseline arm, clean accuracy is 84.5%. At severity 5, matched haze costs 3.1 pp on average (1.9–4.4 across runs), matched cloud 3.9 pp (2.4–5.8), and matched glare 11.3 pp (9.2–17.5). This asymmetry matters for the interpretation of every compound result below: a "haze + glare" compound is in practice a strong degradation with a weak one added, and haze alone is nearly harmless at every severity we tested.

4.2. Compound degradations drop accuracy far below their constituents

At severity 5, haze+glare drops accuracy by 25.7 pp and haze+cloud by 8.6 pp, against largest single drops of 11.3 and 4.0 pp respectively. Across four backbone families the same condition (matched_haze+glare_5) produces drops between −26.1 and −43.3 pp; nine conditions have a mean drop of at least 5 pp with a confidence interval excluding zero in all four families. The compound effect is therefore not a property of one architecture.

4.3. Superadditivity appears at severity 5 only — a threshold, not a gradient

We define superadditivity in accuracy space as

$$E(s) = [\mathrm{acc}(\mathrm{clean}) - \mathrm{acc}(a{+}b, s)] - [\mathrm{acc}(\mathrm{clean}) - \mathrm{acc}(a, s)] + [\mathrm{acc}(\mathrm{clean}) - \mathrm{acc}(b, s)],$$

so that E(s) is the share of the compound drop not explained by the two marginal drops. The additive baseline is the sum of those two marginals, each measured on the same images, same model, same seed, at the same severity, in the same domain as the compound; E(s) is what the sum leaves unexplained. Because a reviewer can reasonably object that any such

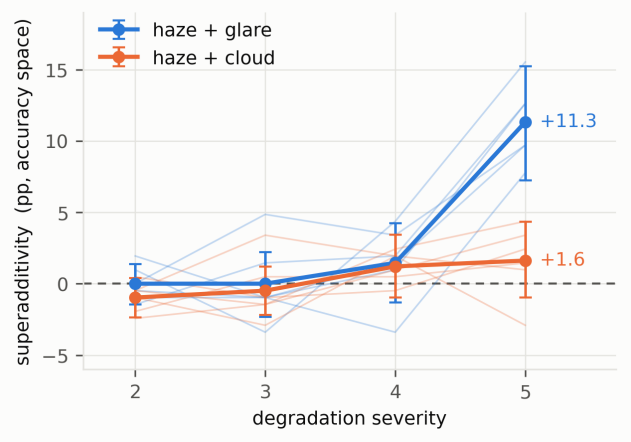


Figure 2. Superadditivity E(s) with 95% cluster-bootstrap intervals (heavy lines) and the six individual (fold, seed) runs (thin lines). Baseline arm.

measure is an artefact of the metric, we report it in accuracy space (the margin-space version appears in the appendix and agrees in sign).

Severity	haze+glare E	95% CI	haze+cloud E	95% CI
2	0.00	[−1.45, +1.38]	−0.97	[−2.36, +0.41]
3	0.00	[−2.34, +2.20]	−0.49	[−2.20, +1.21]
4	+1.46	[−1.30, +4.24]	+1.21	[−0.98, +3.42]
5	+11.33	[+7.22, +15.22]	+1.62	[−0.97, +4.33]

Only haze+glare at severity 5 survives Benjamini–Hochberg correction over the eight estimates ($q < 0.001$); all others have $q \geq 0.48$.

We pre-registered four predictions before running severities 2 and 4. Two are confirmed, one is confirmed and one failed:

- P2 (E at severity 2 covers zero): confirmed.
- P3 (lower CI bound at severity 5 exceeds 2 pp): confirmed ([+7.22]).
- P4 (novelty share increases with severity): confirmed (3.8% → 6.6% → 19.1% → 41.5%).
- P1 (E increases monotonically with severity in at least five of six runs): failed — 2/6. The pooled mean is ordered, but the run-to-run noise (−3.4 to +4.9 pp at severities 2–4) exceeds the difference between adjacent severities.

We report P1's failure rather than redefining the test, because it changes the claim: the data do not support "superadditivity grows with severity". They support a sharper statement: at severities 2–4 the compound effect is additive, and at severity 5 it is not. In all six runs, E(5) exceeds every E(s ≤ 4) of the same run. With no point at severity 4.5, the crossover lies between 4 and 5 and cannot be localised further.

This statement is scoped to one rendering. Under an independent, physically-motivated renderer with severity matched on transmission and saturated-pixel fraction, the severity-5 excess is preserved (+14.6 / +21.8 pp on the frozen baseline, +10.2 / +16.0 pp after retraining under that renderer) but E first becomes large at severity 4 (+4.4 … +9.7 pp). What survives is the excess and its growth with severity; the location of the onset is a property of the renderer (§8.4, reports/PHYSICAL_RENDERER.md).

4.4. Compound failures are new failures

For each compound condition we ask, per image and per seed: is this image misclassified under the compound, and is it also misclassified under any severity of either constituent?

Condition	compound errors	error under no single at any severity	share
haze+glare, s=2	213	8	3.8%
haze+glare, s=3	241	16	6.6%
haze+glare, s=4	298	57	19.1%
haze+glare, s=5	504	209	41.5%
haze+cloud, s=2	207	2	1.0%
haze+cloud, s=3	217	12	5.5%
haze+cloud, s=4	237	17	7.2%
haze+cloud, s=5	292	67	22.9%

Error counts are per (fold, seed) evaluation, summed over the six runs; they share images across seeds and are descriptive, not independent observations. For the severity-3 and severity-5 conditions we additionally compute the overlap with the strongest single condition: 73.4% and 44.4% respectively for haze+glare (reports/COMPOUND_ANALYSIS.md).

At severity 5, four in ten compound errors are errors that no single degradation produces on that image — not even the strongest one, not at any severity. Scope of that statement: it is bounded by the comparison set actually run — three degradation families × five severities × two domains, the same model, the same seed, the same image. It does not claim that no conceivable single degradation could produce them; an untested family, an intermediate severity or a different corruption implementation lies outside the comparison. The phenomenon cannot be explained by saying the compound is simply more severe: there is no single degradation of comparable severity to compare against, since the compound drop (25.7 pp) exceeds the largest single drop (11.3 pp) in all six runs.

This is a behavioural analysis, not a mechanistic one. It does not say why the combination produces new errors; candidate mechanisms — feature loss at the transition between two degradation fields, attention displacement by the glare point source under haze, class-boundary motion — are untested.

4.5. Clean accuracy does not rank models under compound degradation

Two observations in the intervention study (§6) point the same way. The curriculum candidate achieved the highest clean accuracy of any candidate (85.1%) and

the worst compound accuracy (67.3%). The ImageNet-22K initialization improved clean accuracy by +0.73 pp while reducing compound accuracy by −0.71 pp. Neither result is significant on its own, but together they caution against selecting models by clean validation accuracy when deployment is under weather.

4.6. Where the excess comes from: two probes and their limits

The first probe measures the degradation pipeline itself, not the model. Over the 23 matched conditions (three single families × five severities, two compounds × four severities), haze+glare at severity 5 saturates 60.2% of pixels (any channel ≥ 0.99) against 38% for the most saturated single condition (glare_5), and is the only condition that combines high saturation with low contrast (luminance s.d. 0.094). Fitting the condition-level drop as a two-parameter function of saturation fraction and contrast over those 23 conditions gives $R^2 = 0.80$ (coefficients +30.7 / −16.7 pp); adding an interaction term raises it only to 0.814. Two boundaries apply. It is a condition-level relation: at the per-image level the correlation between an image's saturation fraction and its being misclassified is only 0.13 and 0.26 (two seeds). And it is incomplete: haze+cloud at severity 5 saturates nothing, still exceeds the additive prediction, and is the largest residual of the fit (≈6 pp).

The second probe asks whether hard highlight clipping — the mechanism the first probe points at — is what produces the excess. Everything is recomposed as corrupt() does, except that the highlight is rolled off softly above 0.7 instead of clipped at 1.0, and the marginal drops used in the additive prediction are measured in the same re-rendered world. Under clipping the excess at haze+glare_5 is +9.7 pp, under the soft shoulder +6.3 pp (saturation 60.2% → 40.7%). Roughly a third of the excess is attributable to hard clipping; the rest survives without it. This is a single fold and seed on the baseline arm, so it is a directional result, not an estimate.

The third test turns the same reading into a model change and fails: a fourth input channel marking pixels clipped in any channel (zero-initialised mask stem, censored-evidence pooling, +1536 parameters), trained on exactly the baseline recipe and scored on the same 43 paired conditions with per-image McNemar, moves compound accuracy by +0.61 pp ($p = 0.43$, seed 17) and +0.97 pp ($p = 0.27$, seed 29), costs single-degradation accuracy (−0.47 pp, $p = 0.046$; −0.29 pp, $p = 0.41$), and changes sign between seeds on the worst compound condition (−1.9 / +5.8 pp). Telling the network which pixels are censored does not help it.

Together the three probes support one statement only: saturation and contrast are a good description of the severity-5 drop at the level of conditions, and a poor one of which images fail. We do not claim a mechanism for the new errors.

4.7. The interaction is additive over corruption families

Everything above concerns three hand-picked degradations. To ask whether the failure of additivity has structure beyond them, we added six synthetic families (defocus, motion, noise, rain, pixelate, impulse) with continuous levels, calibrated each level so that it alone costs a matched 10 or 20 pp on the frozen baseline, and measured all fifteen pairs over three folds × two seeds.

Raw E conflates interaction with error overlap: two degradations that break the same images cannot add up. Taking the union of the two single-corruption error sets as the independent-failure baseline, delta = e(a+b) − |E_a E_b|, the severity-5 haze+glare excess survives in all six cells (fold0/1/2 × seed17/29; +7.3 to +13.1 pp, mean +9.9), while haze+cloud falls to a mean of −0.7 pp and is positive in only two cells. At severities 2–4 both compounds sit 0.4–2.3 pp below the union baseline. The headline result is therefore not an artefact of overlap, but haze+cloud should not be described as superadditive.

Across the fifteen added pairs, delta ranges from −4.4 to +18.3 pp (ConvNeXt-T). An additive model in per-family main effects, delta(a,b) ≈ m_a + m_b, explains $R^2 = 0.48$–0.90 in all six cells, and $R^2 = 0.42$–0.75 in all six ResNet-50 cells: which pair fails is largely predicted by which families it contains. The family values themselves do not transfer between backbones (impulse contributes +1.7 pp on ConvNeXt and −8.2 pp on ResNet-50), so superadditivity is a property of a (model, corruption) pair rather than of a corruption pair alone.

No single-corruption geometric predictor recovers the ordering — mean feature-shift cosine, per-image displacement alignment, retained high-frequency energy and per-class error correlation all fail ($|\rho| < 0.6$ with unstable sign across cells). Predicting which composition will explode requires measuring part of the composition.

A six-cell replication gate fixed before folds 1–2 were run passed four of five criteria on ConvNeXt-T (the failed one, sub-additivity of two low-pass families, held in 4/6 cells). The ResNet-50 study passes all five but was not pre-registered and is not used to overturn that.

5. Coverage: the excess is an operating point training never visits

The union decomposition of §4.7 says what the excess is not: compound images are not merely harder instances of the errors single degradations already cause. This section asks the complementary question, namely what removes it. The hypothesis is deliberately unglamorous. A model can only learn a joint operating point if training shows it one. Under the project's combined recipe, compounds are a quarter of the batches and are drawn from three families (haze, cloud, glare); the fifteen-pair lattice of §4.7 uses six families that training never mixes. If the excess is an uncovered operating point rather than an intrinsic property of compound degradation, then teaching the model co-occurrence should remove it — and teaching it the same families one at a time should not.

Design. Six synthetic families (defocus, motion, noise, rain, pixelate, impulse), each calibrated to the same marginal cost: the level parameter is swept on a fold's validation split until a single family alone costs 20 pp. Fifteen pairs arise from the six families; three of them (noise+pixelate, defocus+rain, motion+impulse) are held out of every training pool, so nothing below is evaluated on a pair the model could have memorised. Nested balanced pools are drawn over the remaining twelve: c3, c6 and c12 contain 3, 6 and 12 distinct pairs. Every arm — including the baseline, which we re-measured under the same evaluation — gives each sample a clean view with probability 0.5 and a compound otherwise, so compound exposure is held constant and only the diversity of the compound distribution varies. Three folds × two seeds, 40 epochs, ConvNeXt-T, identical to §3.

Result. On the three held-out pairs the baseline excess is +8.2 pp (clean accuracy 84.95%). A pool of three pairs takes it to −1.8 pp (86.25), six pairs to +2.0 (86.08), twelve pairs to +1.4 (86.41). Two things are worth separating. First, the excess is removable: training on a handful of the right operating points turns an 8.2-pp excess into an interaction of the opposite sign, at no clean-accuracy cost. Second, our pre-registered dose-response prediction (P2: monotone improvement in pool size) failed: three pairs is as good as, or better than, twelve, by +3.5 pp (sd 2.2, 6 cells, range −0.5 to +5.3) on held-out pairs. What matters is not how many pairs are in the pool but whether each family is covered by at least one pair — every pool here contains all six, and extra pairs add nothing. Coverage of the family set, not pair diversity, is the operative variable.

What the gain is not. Exposure to the six families is the obvious alternative explanation, so the singles control s6 applies the same six families, at the same 50/50 compound rate, one per draw instead of two. It lands at +7.2 pp (clean 86.73), almost exactly where the baseline is. Against c12 the paired-minus-singles difference is −5.8 pp (range −8.4 to −2.3), and it is negative in 6 of 6 cells; on the twelve pool pairs the difference is −5.1 pp; clean accuracy differs by −0.3 pp. Adding the same information one family at a time does nothing; showing the model the co-occurrence does. The result also holds when the project's own combined recipe is kept in the mixture rather than replaced (§3's recipe, which trains haze, cloud and glare): c12b reaches +4.5 against s6b's +9.7, a paired-minus-singles difference of −5.1 pp (5/6 cells negative, pool pairs −6.0 pp, 6/6).

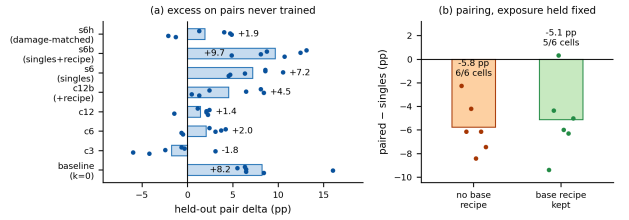


Figure 3. (a) Excess on the three held-out pairs for each arm, one dot per (fold, seed) cell; (b) the paired-minus-singles contrast, in which compound exposure is identical by construction, for the two recipe settings. Negative is better: it means the pair arm has less excess than the same families applied one at a time.

Damage per image is a real confound, and it is part of the effect. A compound image carries roughly twice the damage of a single-family image. We removed this in a third control: s6h applies one family per draw at a level calibrated so that a single corruption costs 40 pp, the same damage per image a pair carries. It reaches +1.9 pp (clean 86.08). Our pre-registered prediction (P12: ≥ +3.0 pp, i.e. that damage-matching would not reproduce the gain) is falsified: matching the damage recovers 5.3 of the 8.9 pp that separates s6 from c3. What survives is smaller but consistent — at equal damage per image, pairs still beat singles by −3.7 pp (range −7.1 to −1.8), 6 of 6 cells, on pairs seen in no training pool. The honest decomposition of the 8.9-pp gain is therefore roughly 60% damage per image and 40% co-occurrence, and the co-occurrence part is the part that survives matching.

Where it stops. Coverage removes the excess only for pairs drawn from families the pools cover. The paper's haze+glare condition is never in any lattice, and its families are absent from the lattice recipe. With the base recipe retained (c12b vs s6b), haze+glare severity-5 accuracy is 56.1% against 57.4%, and the excess 11.0 pp against 12.0 pp — no improvement, as our pre-

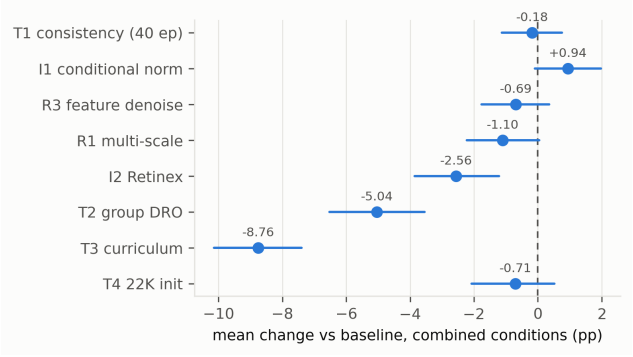


Figure 4. Mean change (pp) against the baseline on the eight combined conditions, with 95% cluster-bootstrap intervals.

registered negative prediction (P10) expected. Coverage transfers to unseen pairs over covered families; it does not transfer to unseen families. This is a boundary of the mechanism and not a defect of the experiment: it says the excess is what an uncovered operating point looks like, and that augmenting over the support of compositions at the families you expect to meet is the operative recipe.

6. Interventions: a pre-registered negative result

We screened seven interventions across three loci plus one initialization control. Every candidate was trained at 40 epochs, three folds, two seeds, with identical data, preprocessing, corruption instances and evaluation. Screening criteria were fixed in advance: a mean gain $\geq +1.5$ pp on the combined conditions, at least two of three folds positive, no clean-accuracy loss beyond 1 pp, and a paired bootstrap interval excluding zero.

Candidate	Locus	Mean Δ	95% CI	Verdict
T1 consistency training	training	−0.18	[−1.13, +0.76]	not sustained
I1 conditional normalization	input	+0.94	[−0.09, +1.98]	below criterion
R3 feature denoising	representation	−0.69	[−1.76, +0.35]	no positive evidence
R1 multi-scale fusion	representation	−1.10	[−2.23, +0.05]	interval covers zero
I2 Retinex preprocessing	input	−2.56	[−3.86, −1.22]	negative
T2 group DRO	training	−5.04	[−6.52, −3.54]	negative
T3 curriculum	training	−8.76	[−10.14, −7.40]	negative
T4 ImageNet-22K init	initialization	−0.71	[−2.07, +0.52]	no advantage demonstrated

Two caveats we insist on. First, R1's interval covers zero: the correct statement is "no detectable gain at +4% parameters and +8.5% latency", not "significant harm". Second, T4 compares two specific weight sets, not "pretraining quality" as a factor.

The budget artefact. T1 showed +1.66 pp [0.63, 2.68] when screened at 24 epochs and −0.18 pp [−1.13, 0.76] at 40 epochs. Both arms improved with the longer budget (84.5 → 85.7 and 84.0 → 85.1 macro-F1), and the candidate's apparent advantage was the difference between two convergence trajectories. Any screening pipeline that compares candidates at a truncated budget inherits this risk; our protocol's full-budget confirmation is what caught it.

Error-flow analysis. Across 6 (fold, seed) pairs × 8 conditions × 206 images, every candidate fixes some baseline errors and breaks more: R1 620 fixed / 728 broken, I2 639/904, T2 608/1103, T3 547/1415. These counts are not independent observations (the same image recurs across conditions and seeds) and are used only to describe direction; formal inference remains the clustered bootstrap.

7. Real imagery: observational only

On 157 FGSC-23 validation images annotated for the presence of real degradation (93 with at least one), the difference in accuracy between degraded and clean groups is −2.3 pp with a seed interval of [−7.8, +4.3]. On 184 two-class FGSRCS development crops the difference is +1.8 pp. Neither supports a claim that real degradation degrades accuracy in this setting, and neither can, because the heavy-degradation stratum contains at most two images. We report these numbers to bound what we claim: our findings concern the synthetic protocol, and we make no claim about their transfer to real weather.

8. Discussion and limitations

8.1. What this work supports

- Under a fixed data set, training recipe and evaluation pipeline, compound degradations are not additive at the highest severity: the joint drop exceeds the sum of marginal drops by 11.3 pp, and exceeds the largest single-degradation drop in every run.
- A substantial share of severe-compound errors (41.5%) cannot be produced by any single degradation, which rules out the "just harder" explanation at the level of behaviour.
- The excess is not intrinsic to compound degradation. A pool of calibrated compound samples over the same six families turns an +8.2 pp held-out excess into −1.8 pp at no clean-accuracy cost; the same families applied one at a time leave it at +7.2 pp (§5). About 40% of that gain survives an arm matched on per-image damage, and only that part transfers to pairs seen in no training pool.
- Seven mechanism-diverse single-point interventions and one initialization control do not beat a strong combined-augmentation baseline. The negative result is pre-registered and reproducible.

8.2. What it does not support

- No performance ceiling. Frequency-domain methods were not tested; deeper backbones, larger data and multi-factor interventions were not tested. We tested seven single-point changes and found no gain; that is all.
- No mechanism. The novelty analysis is behavioural. The saturation/contrast fit of §4.6 is a condition-level association ($R^2 = 0.80$) with weak per-image correlation ($r = 0.13,\ 0.26$) and is not evidence of per-image causation. Both mechanism tests derived from it are negative or partial: removing hard clipping removes about a third of the excess, and a clipped-pixel mask gives no gain. A per-image drift signal does predict failure (AUC 0.78–0.82), but using it as a training weight fails in both directions ($\alpha = +1$: -1.29 and $+0.16$ pp on the six moderate-severity compound conditions; $\alpha = -1$: $+0.65$ and $+2.99$ pp, with the larger gain spread across clean and single degradations rather than compound ones). The diagnostic also does not survive deployment: with the clean reference removed, a test-time cosine distance to the nearest clean class centroid detects errors less well than prediction entropy in all six fold-seed units (-0.020 and -0.034 AUROC on the compound groups). The drift signal is paired by construction, and nothing unpaired reconstructs it. The mechanism behind new errors is unknown.
- No established threshold, and no renderer-independent onset. "Threshold" is an inference from four severity levels, not a localised boundary. Under the shipped renderer E is indistinguishable from its additive prediction at 2, 3 and 4 and large at 5, but with no point at 4.5 the crossover is only bounded to (4, 5); our own pre-registered monotonicity prediction failed (2 of 6 runs). A physically-motivated re-rendering (§8.4) moves the onset to severity 4 while preserving the severity-5 excess, so the supported claim is the descriptive one — an excess that grows with severity — not a law of the severity axis nor a fixed crossover.
- No generalization to real weather. §7 is observational with insufficient heavy-degradation samples.
- No transfer across families. Coverage of co-occurrence removes the excess on pairs drawn from families the training pools contain, and not on haze+glare, whose families are absent from every lattice pool (§5, P10). A deployment recipe must therefore cover the degradations it expects to meet; we have not shown that it can cover degradations it does not model.
- No monotone severity law. Our own pre-registered prediction of monotonicity failed (P1, 2/6); the supported claim is the severity-5 jump.

8.3. Data disclosure

The 236 images in FGSC-23's test split were used by three diagnostic analyses on 2026-10-29 (part occlusion, local recovery, restoration probe). They were not used in training or in any number reported here, but they are no longer an unseen test set, and this paper does not describe them as one. All reported statistics come from folds constructed inside the training split.

8.4. Threats to validity

Single dataset; 618 training images; two seeds; a five-level severity scale with no point between 4 and 5.

Two renderers, not one. The core severity-5 result does not depend on a single corruption implementation. Re-rendering haze with Koschmieder depth-dependent transmission and glare with a specular lobe plus sensor blooming — severity matched to the original on mean transmission and saturated-pixel fraction ($\Delta \le 0.008$ and $\le$ 3e-8) — gives E(5) = +14.6 / +21.8 pp on the frozen baseline, and +10.2 / +16.0 pp after retraining under the new renderer (contrast: +9.7 / +12.6 pp under the original). The excess is therefore a property of the degradation interaction, not of one clipping model. Two limits follow from the same experiment. First, the excess appears from severity 4 under the new renderer, so the onset is renderer-dependent even though the excess is not. Second, matching severity on saturation pins the saturated fraction by construction: the experiment shows that spatial structure matters at fixed saturation, not that saturation is the cause. The physical renderer covers haze+glare only; cloud is unchanged and haze+cloud was not re-rendered. The three remaining single-renderer caveats are the corruption implementations for cloud, the single backbone family used in the physical test, and one fold. The cluster bootstrap accounts for the two seeds per image but not for multiplicity across the full condition grid. The pixel-statistics probe is condition-level and the HDR re-render probe is one fold and one seed, so neither carries an interval. No result here is on an unseen test set (§8.3).

9. Conclusion

Compound weather degradations do something to maritime ship classifiers that single degradations do not: at the highest tested severity their joint effect exceeds the additive prediction by 11.3 pp and exceeds what any single degradation can do, and 41.5% of the resulting errors are errors no single degradation produces. At moderate severities, and under the shipped

renderer, the same conditions are additive; a second, physically-motivated renderer preserves the severity-5 excess but moves its onset down to severity 4, so what we claim is the excess and its growth with severity rather than a fixed crossover. Against the union-of-errors baseline the haze+glare excess survives in all six cells (mean +9.9 pp) while the haze+cloud excess disappears (mean −0.7 pp). Extending to fifteen calibrated pairs, we find that the interaction is additively governed by per-family main effects (R^2 0.42–0.90 across two backbones) but that the per-family values change with the backbone, and that no single-corruption geometric quantity predicts which pair will explode. The excess is removable where it can be covered: a pool of compound samples drawn from the six families brings the held-out excess from +8.2 pp to −1.8 pp in 6 of 6 cells, whereas the same families presented one at a time leave it at +7.2 pp. Matching per-image damage accounts for roughly 60% of that gain and co-occurrence for the remaining 40%, which is the part that generalises to unseen pairs, and none of it generalises to families the pools never contain. Nine interventions failed to improve on a strong combined-augmentation baseline. The protocol, the predictions, the failed pre-registered test and the negative results are released together, so that the negative half of this paper is as checkable as the positive half.

10. Appendix A. Reproduction

Every number in this draft is produced by code in scripts/ from artefacts hashed in reports/snapshot.json (154 runs, code, configs, splits, reports). The probes of §4.6 are reports/SATURATION_PROBE.md (pixel statistics), reports/hdr_render_probe.json (soft-shoulder re-render, script scripts/hdr_render_probe.py) and reports/M1_KILL_TEST.md (mask falsification, script scripts/m1_eval.py, per-image predictions in reports/m1_preds_s17,29.npz). The second renderer of §8.4 is reports/PHYSICAL_RENDERER.md, produced by scripts/physical_eval.py from src/maritime_robust/physical.py and gated by scripts/physical_gate.py; its pre-registration is reports/PREREG_PHYSICAL_RENDERER.md. A full re-evaluation of 12 runs reproduced the published R1 statistic bit-exactly (max absolute difference 0). The pre-registration for the severity 2/4 extension is in reports/PREREG_SEVERITY_2_4.md; the analysis and its comparison against the registered predictions is in reports/SEVERITY_CURVE.md. The error-overlap decomposition of §4.7 is reports/COMPOSITION.md §2, from scripts/overlap_main_grid.py; the fifteen-pair composition study and its replication gate are the rest of reports/COMPOSITION.md, from scripts/composition_calibrate,pairs,interaction,predictors,aggregate with the gate pre-registered in reports/PREREG_COMPOSITION_REPLICATION.md. All composition measurements are frozen-model inference; the only training added is six ResNet-50 clean baselines.

11. Appendix B. Margin-space superadditivity

The original analysis measured superadditivity in margin space (difference between compound accuracy and the additive expectation of margin losses): haze+glare extra margin loss −0.152 … −0.032, haze+cloud −0.144 … +0.009, across four backbone families. It agrees in sign with the accuracy-space result reported in §4.3 and is not used for inference here.

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